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E-Learning Material

Fundamentals of Mathematics

(25VBT0C102)

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Module: 03

Different Calculus & Applications

Module Information

Field	Details
Course Title	Fundamentals of Mathematics
Course Code	25VBT0C102
Module Number	Module 3 of 5
Module Title	Differential Calculus and Applications

Module Overview

Calculus is the mathematics of change, and differentiation is its most fundamental tool. While algebra deals with static quantities and geometry describes fixed shapes, differential calculus captures motion, growth, curvature, and optimisation — the dynamic aspects of the world that algebra alone cannot express. Isaac Newton and Gottfried Wilhelm Leibniz independently developed calculus in the 17th century to solve physical problems involving motion and tangents, and the notation we use today (dy/dx) comes directly from Leibniz.

This module begins at the foundation: the concept of continuity. Before we can differentiate a function, we must ask whether it is well-behaved at a point — whether small changes in input produce small changes in output, or whether the function jumps, breaks, or oscillates wildly. Continuity at a point and over intervals establishes the minimal requirement for a function to be differentiable.

From continuity, you will move to derivatives — the core operation of differential calculus. The derivative of a function at a point measures the instantaneous rate of change: how fast the output is changing at that instant. We begin with derivatives of the fundamental function types — algebraic (polynomial, rational, radical), trigonometric (sin, cos, tan, and their inverses), exponential, and logarithmic — building a complete toolkit of differentiation rules.

Three powerful rules extend the basic derivatives to complex functions: the Chain Rule for composite functions (functions of functions), the rules for implicit differentiation when variables cannot be separated, and the formulas for parametric differentiation when both x and y are given as functions of a third parameter t. Together these cover virtually every function encountered in engineering and science.

Higher-order derivatives form the next step. The second derivative measures the rate of change of the rate of change — it is the concavity of a curve. The n-th derivative is defined recursively. The Leibnitz Theorem provides a beautiful closed-form formula for the n-th derivative of a product of two functions, generalising the product rule to any order.

The module then turns to elegant geometric applications. The angle between two curves at their point of intersection can be computed using derivatives — specifically the tangent slopes. The pedal equation connects the distance from the origin to a tangent of a curve with the radial distance, providing a powerful invariant for curve families.

Finally, polar coordinates provide an alternative coordinate system where curves are described by their distance from a fixed point (the pole) as a function of angle. Polar curves include spirals, roses, cardioids, and lemniscates — shapes that are very natural in polar coordinates but complicated in Cartesian form. Differentiation in polar coordinates, including arc length and the angle between the radius and the tangent, completes the module.

Learning Objectives

By the end of this module, you will be able to:

- Understand the concept of continuity of a function at a point and over an interval and state the conditions required for differentiability.
- Apply differentiation rules — including the chain rule, product rule, quotient rule, and logarithmic differentiation — to algebraic, trigonometric, exponential, logarithmic, parametric, and inverse trigonometric functions.
- Compute second-order and n-th order derivatives, and apply the Leibnitz Theorem to find the n-th derivative of a product of two functions.
- Analyse the geometric relationship between two intersecting curves by computing the angle of intersection using derivatives.
- Differentiate polar curves and apply polar differentiation to compute the angle between the radius vector and the tangent, and to derive and interpret the pedal equation.

Learning Outcomes

After completing this module, you will be able to:

- Test a function for continuity at a given point and identify removable, jump, and infinite discontinuities.
- Differentiate composite, implicit, parametric, and inverse trigonometric functions using appropriate rules and verify results.
- Apply the Leibnitz Theorem to compute the n-th derivative of products such as $x^n e^x$, $x^n \sin(x)$, or $x^n \log(x)$.
- Find the angle of intersection of two given curves at their points of intersection.
- Convert between Cartesian and polar forms and compute derivatives and pedal equations of standard polar curves.

Table of Topics

Sub-Unit	Topic	Description
3.1	Continuity of Functions	Definition at a point, left and right limits, types of discontinuities, continuity over intervals
3.2	Concept of the Derivative	Rate of change, first principles (limit definition), geometric meaning as slope of tangent
3.3	Derivatives of Algebraic Functions	Power rule, sum/difference, constant multiple, polynomial and rational functions
3.4	Derivatives of Trigonometric Functions	d/dx(sin x), cos x, tan x, cot x, sec x, cosec x; standard results and proofs
3.5	Derivatives of Exponential and Logarithmic Functions	$d/dx(e^x)$, a^x, ln x, $\log_a x$; logarithmic differentiation technique
3.6	Chain Rule and Composite Functions	Chain rule statement, nested compositions, applying to complex functions
3.7	Product Rule and Quotient Rule	$d/dx(uv) = uv' + vu'$, $d/dx(u/v) = (vu' - uv')/v^2$, applications
3.8	Derivatives of Inverse Trigonometric Functions	$d/dx(\sin^{-1}x)$, $\cos^{-1}x$, $\tan^{-1}x$, $\cot^{-1}x$, $\sec^{-1}x$, $\operatorname{cosec}^{-1}x$
3.9	Implicit Differentiation	Differentiating both sides w.r.t. x when y cannot be isolated
3.10	Parametric Differentiation	$dy/dx = (dy/dt)/(dx/dt)$; second derivative in parametric form
3.11	Second-Order Derivatives	d^2y/dx^2, concavity, inflection points; second derivative in parametric form
3.12	nth-Order Derivatives and Leibnitz Theorem	Recursive definition, standard nth derivatives, Leibnitz formula for $D^n(uv)$
3.13	Angle Between Two Curves	Using slope formula $m = \tan\theta$, angle of intersection, orthogonal curves
3.14	Polar Coordinates and Polar Curves	Polar equation $r = f(\theta)$, common curves, conversion, sketching
3.15	Differentiation in Polar Coordinates and Pedal Equation	$\tan\varphi = r/(dr/d\theta)$, pedal equation $p = f(r)$

3.1 Continuity of Functions

Before we can discuss the derivative of a function, we must first understand continuity. Continuity is the formal mathematical expression of the intuitive idea that a function has no "breaks", "jumps", or "holes" — that its graph can be drawn without lifting the pen. While this intuition is helpful, the precise definition using limits is essential for rigorous analysis.

Limit of a Function at a Point

The limit of f(x) as x approaches a, written $\lim_{x\to a} f(x) = L$, means that f(x) can be made arbitrarily close to L by taking x sufficiently close to a (but not equal to a). The limit captures the behaviour of f near the point a, independent of the value f(a) itself.

For a limit to exist at a point, the left-hand limit and right-hand limit must both exist and be equal:

Condition for limit to exist: $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$

Definition of Continuity at a Point

A function f(x) is continuous at a point x = a if all three of the following conditions are satisfied simultaneously:

- Condition 1 — Defined: f(a) exists (the function has a value at a).
- Condition 2 — Limit exists: $\lim_{x\to a} f(x)$ exists (the left and right limits are equal).
- Condition 3 — Agreement: $\lim_{x\to a} f(x) = f(a)$ (the limit equals the function value).

If any one of these three conditions fails, the function is discontinuous at x = a.

Types of Discontinuities

Type	Description	Example / Behaviour
Removable Discontinuity	Limit exists, but f(a) is undefined or ≠ limit. The "hole" can be filled by redefining f(a).	$f(x) = (x^2-1)/(x-1)$ at $x=1$: limit = 2, but $f(1)$ undefined
Jump Discontinuity	Left and right limits both exist but are not equal. The function "jumps" from one value to another.	Floor function $\lfloor x \rfloor$ at $x=1$: left limit=0, right limit=1
Infinite Discontinuity	The function approaches $\pm\infty$ as $x \to a$. A vertical asymptote exists at $x = a$.	$f(x) = 1/x$ at $x=0$: limit $\to \pm\infty$
Oscillatory Discontinuity	The function oscillates infinitely rapidly near a. Neither left nor right limit exists.	$f(x) = \sin(1/x)$ at $x=0$: no limit exists

Continuity Over an Interval

A function is continuous on an open interval (a, b) if it is continuous at every point in the interval. A function is continuous on a closed interval [a, b] if it is continuous on the open interval (a, b) and the one-sided limits at the endpoints equal the function values: $\lim[x\to a^+]$ f(x) = f(a) and $\lim[x\to b^-]$ f(x) = f(b).

Continuity and Differentiability

There is an important one-way relationship: differentiability implies continuity, but continuity does not imply differentiability. If a function is differentiable at a point, it must be continuous there. However, a continuous function may fail to be differentiable — for example, f(x) = |x| is continuous at x = 0 but has a sharp corner (not differentiable there).

📝 Worked Example: Testing Continuity

Test whether $f(x) = (x^2 - 4)/(x - 2)$ for $x \neq 2$, and f(2) = 4, is continuous at x = 2.

Step 1: Check if f(2) is defined: f(2) = 4. ✓

Step 2: Compute the limit: $\lim[x\to2]\ (x^2-4)/(x-2) = \lim[x\to2]\ (x+2)(x-2)/(x-2) = \lim[x\to2]\ (x+2) = 4$

Step 3: Compare: $\lim[x\to2]$ f(x) = 4 = f(2). All three conditions satisfied. ✓

Therefore f(x) is continuous at x = 2. (This is a removable discontinuity that has been "repaired" by defining f(2) = 4.)

3.2 The Concept of the Derivative

The derivative is the mathematical formalisation of the idea of instantaneous rate of change. Consider a car travelling along a road: its average speed over a time interval is distance/time. But what is its speed at a single instant — at precisely 3:00:00 PM? To answer this, we let the time interval shrink to zero and take the limit. This limiting process gives the instantaneous rate of change, and that limit is the derivative.

Derivative from First Principles

Let $y = f(x)$ be a function defined in an open interval containing x. The derivative of f at x, denoted $f'(x)$ or dy/dx, is defined as:

First principles definition: $f'(x) = \lim_{h \to 0} [f(x+h) - f(x)] / h$

This limit, if it exists, is called the derivative of f at x. The notation dy/dx (Leibniz notation) and $f'(x)$ (Lagrange notation) are both standard and widely used. The expression $[f(x+h) - f(x)] / h$ is called the difference quotient, and it represents the slope of the secant line through $(x, f(x))$ and $(x+h, f(x+h))$.

Geometric Interpretation

Geometrically, the derivative $f'(a)$ is the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$. As $h \to 0$, the secant line (connecting two nearby points on the curve) rotates and converges to the tangent line. The tangent line at $(a, f(a))$ has equation:

Equation of tangent line: $y - f(a) = f'(a) \cdot (x - a)$

📝 Worked Example: Derivative from First Principles

Find the derivative of $f(x) = x^2$ from first principles.

Step 1: Apply the definition:

$f'(x) = \lim_{h\to 0} [(x+h)^2 - x^2] / h$

Step 2: Expand the numerator:

$= \lim_{h\to 0} [x^2 + 2xh + h^2 - x^2] / h \;=\; \lim_{h\to 0} [2xh + h^2] / h$

Step 3: Factor out h:

$= \lim_{h\to 0} h(2x + h) / h \;=\; \lim_{h\to 0} (2x + h) \;=\; 2x$

Therefore $f'(x) = 2x$. At $x = 3$, the slope of the tangent to $y = x^2$ is $f'(3) = 6$.

💡 Did You Know?

The notation dy/dx was invented by Gottfried Wilhelm Leibniz in the 1670s and remains universally standard today. Isaac Newton independently developed calculus using the notation $\dot{x}$ (x-dot for the derivative with respect to time), still used in physics. The Indian mathematician Bhāskara II described instantaneous velocity (tātkālika-gati) in the 12th century in his work Siddhānta Śiromaṇi, anticipating the conceptual basis of the derivative by five centuries. Modern applications of derivatives in India include ISRO's trajectory optimisation, the pricing algorithms of NSE (National Stock Exchange), and aerodynamic models developed at NAL (National Aerospace Laboratories) in Bengaluru.

3.3 Derivatives of Algebraic Functions

Rather than applying first principles to every function, we build a library of differentiation rules that allow rapid computation. The most fundamental rules apply to algebraic functions — polynomials, rational functions, and radical expressions.

Standard Differentiation Rules

Rule	Formula	Example
Constant Rule	$d/dx(c) = 0$	d/dx(7) = 0
Power Rule	$d/dx(x^n) = nx^{n-1}$ $(n \in \mathbb{R})$	$d/dx(x^5) = 5x^4$; d/dx(x^(1/2)) = (1/2)x^(−1/2)
Constant Multiple	$d/dx(c \cdot f) = c \cdot f'$	$d/dx(3x^4) = 12x^3$
Sum / Difference	$d/dx(f \pm g) = f' \pm g'$	$d/dx(x^3 + 2x) = 3x^2 + 2$
Product Rule	$d/dx(u \cdot v) = u'v + uv'$	d/dx(x·sin x) = sin x + x·cos x
Quotient Rule	$d/dx(u/v) = (u'v - uv') / v^2$	$d/dx(x^2/\sin x) = (2x \cdot \sin x - x^2 \cdot \cos x)/\sin^2 x$

Worked Example: Differentiating Polynomial and Rational Functions

Differentiate: (i) $f(x) = 3x^4 - 5x^3 + 7x - 2$ (ii) $g(x) = (x^2 + 1)/(x - 1)$

Step 1: $f(x) = 3x^4 - 5x^3 + 7x - 2$ (apply power rule term-by-term):

$f'(x) = 12x^3 - 15x^2 + 7$

Step 2: $g(x) = (x^2 + 1)/(x - 1)$. Apply quotient rule: $u = x^2 + 1$, $v = x - 1$.

$u' = 2x,\quad v' = 1$

$g'(x) = (u'v - uv')/v^2 = [(2x)(x-1) - (x^2+1)(1)] / (x-1)^2$

$= [2x^2 - 2x - x^2 - 1] / (x-1)^2$

$= (x^2 - 2x - 1) / (x - 1)^2$

3.4 Derivatives of Trigonometric Functions

The six trigonometric functions each have well-defined derivatives that can be derived from first principles using limit results. These derivatives are among the most important in calculus because trigonometric functions appear in almost every physical model involving oscillation, waves, or rotation.

Standard Trigonometric Derivatives

Function	Derivative	Key Observation
d/dx(sin x)	cos x	The derivative of sin is cos
d/dx(cos x)	−sin x	Negative sign — slope of cosine is negative sine
d/dx(tan x)	$\sec^2 x$	Equal to $1 + \tan^2 x$; useful in integration
d/dx(cot x)	$-\text{cosec}^2 x$	Negative of $\text{cosec}^2 x$
d/dx(sec x)	sec x · tan x	Product of sec and tan

d/dx(cosec x)	$-\text{cosec } x \cdot \cot x$	Negative; note product form

Worked Example: Differentiating Trigonometric Functions

Find the derivatives of: (i) $y = x^2 \sin x$ (ii) $y = \tan x / (1 + \sec x)$

Step 1: $y = x^2 \sin x$. Apply product rule: $u = x^2$, $v = \sin x$.

$u' = 2x$, $v' = \cos x$

$dy/dx = u'v + uv' = 2x \cdot \sin x + x^2 \cdot \cos x$

Step 2: $y = \tan x / (1 + \sec x)$. Apply quotient rule: $u = \tan x$, $v = 1 + \sec x$.

$u' = \sec^2 x$, $v' = \sec x \cdot \tan x$

$dy/dx = [\sec^2 x \cdot (1 + \sec x) - \tan x \cdot \sec x \cdot \tan x] / (1 + \sec x)^2$

$= [\sec^2 x + \sec^3 x - \sec x \cdot \tan^2 x] / (1 + \sec x)^2$

$= \sec x[\sec x + \sec^2 x - \tan^2 x] / (1 + \sec x)^2$

Since $\sec^2 x - \tan^2 x = 1$: $= \sec x[\sec x + 1] / (1 + \sec x)^2 = \sec x / (1 + \sec x)$

3.5 Derivatives of Exponential and Logarithmic Functions

Exponential and logarithmic functions are ubiquitous in modelling real phenomena: radioactive decay, compound interest, population growth, information entropy, and signal processing all involve these functions. Their derivatives are surprisingly clean and elegant.

Standard Results

$d/dx(e^x) = e^x$ — the exponential function is its own derivative!

$d/dx(a^x) = a^x \cdot \ln a$ (for any base $a > 0$, $a \neq 1$)

$$\frac{d}{dx}(\ln x) = 1/x \quad (\text{for } x > 0)$$

$$\frac{d}{dx}(\log_a x) = 1/(x \cdot \ln a) \quad (\text{change of base formula applied})$$

The fact that e^x is its own derivative is what makes Euler's number $e \approx 2.71828...$ uniquely special. No other base has this property. This is precisely why all exponential models in calculus are naturally expressed base e.

Logarithmic Differentiation

For functions of the form $y = [u(x)]^{v(x)}$ — where both the base and the exponent contain x — neither the power rule nor the exponential rule directly applies. The technique of logarithmic differentiation handles this: take the natural log of both sides, differentiate implicitly, then solve for dy/dx.

The steps are: (1) Write $\ln y = v(x) \cdot \ln[u(x)]$. (2) Differentiate both sides w.r.t. x: $(1/y)(dy/dx) = v'(x)\cdot\ln[u(x)] + v(x)\cdot u'(x)/u(x)$. (3) Multiply both sides by y to get dy/dx.

📝 Worked Example: Logarithmic Differentiation

Find dy/dx for: (i) $y = x^x$ (ii) $y = (\sin x)^{\cos x}$

Step 1: $y = x^x$. Take ln of both sides: $\ln y = x \cdot \ln x$

Differentiate: $(1/y)(dy/dx) = \ln x + x\cdot(1/x) = \ln x + 1$

$dy/dx = y(\ln x + 1) = x^x(\ln x + 1)$

Step 2: $y = (\sin x)^{\cos x}$. Take ln: $\ln y = \cos x \cdot \ln(\sin x)$

$(1/y)(dy/dx) = -\sin x \cdot \ln(\sin x) + \cos x \cdot (\cos x / \sin x)$

$\qquad = -\sin x \cdot \ln(\sin x) + \cos^2 x / \sin x$

$dy/dx = (\sin x)^{\cos x} \cdot [\cos^2 x/\sin x - \sin x \cdot \ln(\sin x)]$

✓ Let's Check

Q.No	Type	Question
1	MCQ	The derivative of e^x is:
2	MCQ	If $y = x^3 + 5x - 2$, then dy/dx at $x = 1$ is:
3	Fill in Blank	The derivative of ln x is ______.
4	Fill in Blank	A function f(x) is continuous at x = a if lim[x→a] f(x) = ______.
5	True/False	If f(x) is differentiable at x = a, then it must also be continuous at x = a.
6	True/False	The derivative of a^x (for $a > 0$, $a \neq 1$) is a^(x−1).

Answers:

Q.No	Answer
1	c) e^x — the natural exponential function is its own derivative.
2	b) 8 — $dy/dx = 3x^2 + 5$; at x=1: 3(1) + 5 = 8.
3	1/x
4	f(a)
5	True — differentiability implies continuity. The derivative limit cannot exist if the function is discontinuous.
6	False — $d/dx(a^x) = a^x \cdot \ln a$, NOT a^(x−1). The power rule requires a constant exponent, not a variable.

3.6 he Chain Rule and Composite Functions

Many functions in mathematics and applications are built by composing simpler functions: $\sin(x^2)$, e^(3x+1), ln(cos x), $(1 + x^2)$^(1/3). The Chain Rule is the rule that allows us to differentiate any composite function. It is arguably the single most important differentiation rule, because virtually every real-world function is a composition of simpler ones.

If y = f(g(x)) — a function of a function — then the derivative of y with respect to x is the product of two derivatives: the outer function's derivative (evaluated at the inner function) times the inner function's derivative:

Chain Rule: dy/dx = f'(g(x)) · g'(x)

In Leibniz notation, if y = f(u) and u = g(x), then:

Chain Rule — Leibniz form: dy/dx = (dy/du) · (du/dx)

The Leibniz form has an elegant "cancellation" interpretation: the du terms appear to cancel, giving dy/dx. While this is not a rigorous proof, it makes the rule easy to remember.

Generalised Power Rule

A particularly important application of the chain rule is the generalised power rule, which handles any function raised to a power:

Generalised Power Rule: $\frac{d}{dx}[g(x)]^n = n\cdot[g(x)]^{n-1} \cdot g'(x)$

Worked Example: Chain Rule Applications

Differentiate: (i) $y = \sin(3x^2 + 1)$ (ii) $y = e^{(x^2 - 2x)}$ (iii) $y = (1 + \tan x)^5$

Step 1: $y = \sin(3x^2 + 1)$. Outer: $\sin(u)$, $u = 3x^2+1$.

$dy/dx = \cos(3x^2 + 1) \cdot d/dx(3x^2 + 1) = \cos(3x^2 + 1) \cdot 6x = 6x\cdot\cos(3x^2 + 1)$

Step 2: $y = e^{(x^2 - 2x)}$. Outer: e^u, $u = x^2-2x$.

$dy/dx = e^{(x^2 - 2x)} \cdot d/dx(x^2 - 2x) = e^{(x^2 - 2x)} \cdot (2x - 2) = (2x-2)\cdot e^{(x^2-2x)}$

Step 3: $y = (1 + \tan x)^5$. Apply generalised power rule: $n=5$, $g(x) = 1+\tan x$.

$dy/dx = 5(1 + \tan x)^4 \cdot \sec^2 x$

Worked Example: Chain Rule — Multi-Layer Composition

Find dy/dx for $y = \ln(\sin(e^x))$.

Step 1: Identify layers: outermost is $\ln(\cdot)$, middle is $\sin(\cdot)$, innermost is e^x.

Step 2: Apply chain rule layer by layer:

$dy/dx = [1/\sin(e^x)] \cdot d/dx[\sin(e^x)]$

$= [1/\sin(e^x)] \cdot \cos(e^x) \cdot d/dx(e^x)$

$= [1/\sin(e^x)] \cdot \cos(e^x) \cdot e^x$

$= e^x \cdot \cos(e^x) / \sin(e^x) = e^x \cdot \cot(e^x)$

3.7 Product Rule and Quotient Rule — Further Applications

The product and quotient rules were introduced in Sub-unit 3.3. Here we develop deeper facility with these rules through more complex applications and explore how they interact with the chain rule. In practice, most differentiation problems require simultaneously applying the chain rule, product rule, and/or quotient rule.

📝 Worked Example: Combined Rules — Product with Chain

Differentiate: (i) $y = x^2 \cdot e^{(3x)}$ (ii) $y = \sqrt{x} \cdot \ln x$ (iii) $y = \sin^2 x \cdot \cos 3x$

Step 1: $y = x^2 \cdot e^{(3x)}$. Product rule: $u = x^2$, $v = e^{(3x)}$.

$u' = 2x$, $v' = 3e^{(3x)}$ (chain rule on v)

$dy/dx = 2x \cdot e^{(3x)} + x^2 \cdot 3e^{(3x)} = e^{(3x)}(2x + 3x^2) = xe^{(3x)}(2 + 3x)$

Step 2: $y = \sqrt{x} \cdot \ln x = x^{(1/2)} \cdot \ln x$. Product rule: $u = x^{(1/2)}$, $v = \ln x$.

$u' = (1/2)x^{(-1/2)}$, $v' = 1/x$

$dy/dx = (1/2)x^{(-1/2)} \cdot \ln x + x^{(1/2)} \cdot (1/x) = (\ln x)/(2\sqrt{x}) + 1/\sqrt{x}$

$= (\ln x + 2) / (2\sqrt{x})$

Step 3: $y = \sin^2 x \cdot \cos 3x$. Chain rule gives: $d/dx(\sin^2 x) = 2 \sin x \cdot \cos x = \sin 2x$

$dy/dx = \sin 2x \cdot \cos 3x + \sin^2 x \cdot (-3 \sin 3x) = \sin 2x \cdot \cos 3x - 3 \sin^2 x \cdot \sin 3x$

3.8 Derivatives of Inverse Trigonometric Functions

The inverse trigonometric functions ($\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, etc.) arise naturally when we need to find angles from ratios. Their derivatives involve algebraic expressions containing square roots and are derived using implicit differentiation together with the Pythagorean identity. These derivatives appear extensively in integration problems (as antiderivatives).

Standard Results

Function	Derivative
$d/dx(\sin^{-1} x)$	$1 / \sqrt{(1 - x^2)}$ valid for $\lvert x\rvert < 1$
$d/dx(\cos^{-1} x)$	$-1 / \sqrt{(1 - x^2)}$ valid for $\lvert x\rvert < 1$
$d/dx(\tan^{-1} x)$	$1 / (1 + x^2)$ valid for all x
$d/dx(\cot^{-1} x)$	$-1 / (1 + x^2)$ valid for all x
$d/dx(\sec^{-1} x)$	$1 / (\lvert x\rvert\sqrt{(x^2 - 1)})$ valid for $\lvert x\rvert > 1$
$d/dx(\mathrm{cosec}^{-1} x)$	$-1 / (\lvert x\rvert\sqrt{(x^2 - 1)})$ valid for $\lvert x\rvert > 1$

Note that $d/dx(\sin^{-1}x)$ and $d/dx(\cos^{-1}x)$ are negatives of each other; similarly, $d/dx(\tan^{-1}x)$ and $d/dx(\cot^{-1}x)$ are negatives of each other. This means $\sin^{-1}x + \cos^{-1}x = \pi/2$ (constant), and $\tan^{-1}x + \cot^{-1}x = \pi/2$ (constant), both of which imply their derivatives sum to zero.

Worked Example: Derivatives of Inverse Trigonometric Functions

Differentiate: (i) $y = \sin^{-1}(2x)$ (ii) $y = \tan^{-1}(x/a)$ (iii) $y = x \cdot \cos^{-1}x$

Step 1: $y = \sin^{-1}(2x)$. Apply chain rule: outer = $\sin^{-1}(u)$, $u = 2x$.

$dy/dx = 1/\sqrt{(1-(2x)^2)} \cdot 2 = 2/\sqrt{(1 - 4x^2)}$

Step 2: $y = \tan^{-1}(x/a)$. Apply chain rule: outer = $\tan^{-1}(u)$, $u = x/a$.

$dy/dx = 1/(1+(x/a)^2) \cdot (1/a) = 1/[(a^2+x^2)/a^2] \cdot (1/a) = a/(a^2+x^2)$

Step 3: $y = x \cdot \cos^{-1}x$. Apply product rule: $u = x$, $v = \cos^{-1}x$.

$u' = 1$, $v' = -1/\sqrt{(1-x^2)}$

$dy/dx = \cos^{-1}x + x\cdot(-1/\sqrt{(1-x^2)}) = \cos^{-1}x - x/\sqrt{(1-x^2)}$

Analysing Misconceptions: Chain Rule: 'Differentiate from outside in'

✕ MYTH: When applying the chain rule to $\sin(3x^2)$, we get $\cos(3x^2)$ and forget the inner derivative, giving just $\cos(3x^2)$.

✓ REALITY: The chain rule requires multiplying by the derivative of the inner function. For $y = \sin(3x^2)$, the outer derivative is $\cos(3x^2)$ and the inner derivative is $6x$. The correct answer is $dy/dx = 6x\cdot\cos(3x^2)$. Missing the inner derivative is the single most common error in differentiation. Always ask: 'Did I differentiate the inner function and multiply?' This applies to every chain rule application without exception.

3.9 Implicit Differentiation

Up to this point, all functions have been in explicit form: y expressed directly as a function of x. However, many important curves and surfaces are defined by equations of the form $F(x, y) = 0$ where y cannot be isolated algebraically, or where isolation would produce multiple branches. Examples include circles ($x^2 + y^2 = r^2$), ellipses, and folium curves. Implicit differentiation allows us to find dy/dx without ever solving for y explicitly.

The key principle is: differentiate both sides of the equation with respect to x, treating y as an implicit function of x. Whenever y appears, its derivative with respect to x is dy/dx — apply the chain rule. Then collect all dy/dx terms on one side and solve.

Worked Example: Implicit Differentiation

Find dy/dx for: (i) $x^2 + y^2 = 25$ (ii) $x^3 + y^3 = 3xy$ (iii) $e^{(xy)} = x + y$

Step 1: $x^2 + y^2 = 25$. Differentiate both sides w.r.t. x:

$2x + 2y(dy/dx) = 0$

$dy/dx = -2x/(2y) = -x/y$

(This is the slope of the circle of radius 5 at the point (x, y).)

Step 2: $x^3 + y^3 = 3xy$. Differentiate both sides:

$3x^2 + 3y^2(dy/dx) = 3[y + x(dy/dx)]$ (product rule on right side)

$3y^2(dy/dx) - 3x(dy/dx) = 3y - 3x^2$

$3(y^2 - x)(dy/dx) = 3(y - x^2)$

$dy/dx = (y - x^2)/(y^2 - x)$

Step 3: $e^{(xy)} = x + y$. Differentiate left side using chain rule:

$e^{(xy)} \cdot [y + x(dy/dx)] = 1 + dy/dx$

$ye^{(xy)} + xe^{(xy)}(dy/dx) = 1 + dy/dx$

$xe^{(xy)}(dy/dx) - dy/dx = 1 - ye^{(xy)}$

$dy/dx \cdot (xe^{(xy)} - 1) = 1 - ye^{(xy)}$

$dy/dx = (1 - ye^{(xy)}) / (xe^{(xy)} - 1)$

3.10 Parametric Differentiation

In parametric form, both x and y are expressed as functions of a third variable — the parameter, commonly denoted t or θ. Parametric equations are natural when describing motion (x(t) = position at time t) or when curves are more easily described by their generating process than by an explicit y = f(x) relationship. The unit circle, for instance, is most naturally written as x = cos t, y = sin t.

For parametric curves x = f(t), y = g(t), the derivative dy/dx is found using the chain rule in the form:

Parametric derivative: $dy/dx = (dy/dt) / (dx/dt) = g'(t) / f'(t)$ [provided $dx/dt \neq 0$]

Second Derivative in Parametric Form

The second derivative d^2y/dx^2 requires an additional application of the chain rule. The key insight is that $d^2y/dx^2 = d/dx(dy/dx)$ is not $(d^2y/dt^2)/(d^2x/dt^2)$. Instead:

Second derivative parametric: $d^2y/dx^2 = d/dx(dy/dx) = [d/dt(dy/dx)] / (dx/dt)$

📝 Worked Example: Parametric Differentiation

Find dy/dx and d^2y/dx^2 for the parametric curve: x = t − sin t, y = 1 − cos t (the cycloid).

Step 1: Compute dx/dt and dy/dt:

$dx/dt = 1 - \cos t$

$dy/dt = \sin t$

Step 2: First derivative:

$dy/dx = (dy/dt)/(dx/dt) = \sin t / (1 - \cos t)$

Using trig identity: $\sin t = 2 \sin(t/2)\cos(t/2)$ and $1-\cos t = 2\sin^2(t/2)$:

$dy/dx = [2 \sin(t/2)\cos(t/2)] / [2\sin^2(t/2)] = \cos(t/2)/\sin(t/2) = \cot(t/2)$

Step 3: Second derivative: find $d/dt(dy/dx)$ first.

$d/dt[\cot(t/2)] = -\operatorname{cosec}^2(t/2) \cdot (1/2) = -(1/2)\operatorname{cosec}^2(t/2)$

Step 4: Divide by dx/dt:

$d^2y/dx^2 = [-(1/2)\operatorname{cosec}^2(t/2)] / [1 - \cos t] = [-(1/2)\operatorname{cosec}^2(t/2)] / [2\sin^2(t/2)]$

$\quad = -1 / [4\sin^4(t/2)]$

📝 Worked Example: Parametric Curve — Ellipse

For $x = a \cdot \cos t$, $y = b \cdot \sin t$ (ellipse), find dy/dx and the equation of the tangent at parameter t.

Step 1: $dx/dt = -a \cdot \sin t$, $\quad dy/dt = b \cdot \cos t$

Step 2: $dy/dx = b \cdot \cos t / (-a \cdot \sin t) = -(b/a) \cdot \cot t$

Step 3: At point $(a \cos t, b \sin t)$, tangent equation:

$y - b \cdot \sin t = -(b/a) \cdot \cot t \cdot (x - a \cdot \cos t)$

Simplifying: $bx \cdot \cos t + ay \cdot \sin t = ab$ (standard tangent to ellipse)

✓ Let's Check

Q.No	Type	Question
1	MCQ	For $y = f(g(x))$, the chain rule gives dy/dx as:
2	MCQ	For parametric equations $x = t^2$, $y = t^3$, the value of dy/dx at $t = 2$ is:
3	Fill in Blank	The derivative of $\sin^{-1}(x)$ is ______.
4	Fill in Blank	In implicit differentiation of $x^2 + y^2 = 25$, the term $2y$ must be multiplied by ______ when differentiating y^2 w.r.t. x.
5	True/False	For parametric form, $d^2y/dx^2 = (d^2y/dt^2)/(d^2x/dt^2)$.
6	True/False	Logarithmic differentiation is used when both the base and exponent of a function contain the variable x.

Answers:

Q.No	Answer
1	b) $f'(g(x)) \cdot g'(x)$ — the chain rule multiplies the outer derivative by the inner derivative.
2	c) 3 — $dx/dt = 2t$, $dy/dt = 3t^2$; $dy/dx = 3t^2/2t = 3t/2$. At $t=2$: $dy/dx = 3(2)/2 = 3$.
3	$1/\sqrt{(1-x^2)}$
4	dy/dx

5	False — $d^2y/dx^2 = [d/dt(dy/dx)] / (dx/dt)$. The formula is NOT the ratio of second derivatives.
6	True — functions like x^x or $(\sin x)^{\wedge}(\cos x)$ require logarithmic differentiation.

3.11 Second-Order Derivatives

The second derivative of a function $y = f(x)$, denoted $f''(x)$ or d^2y/dx^2, is the derivative of the derivative. While the first derivative measures the rate of change of y, the second derivative measures the rate of change of the rate of change — it captures how quickly the slope itself is changing. This is a measure of curvature or concavity.

Concavity and the Second Derivative

The second derivative gives the concavity of a curve at any point:

- $f''(x) > 0$ at a point: the function is concave upward (bends upward like a cup) — the slope is increasing.
- $f''(x) < 0$ at a point: the function is concave downward (bends downward like a cap) — the slope is decreasing.
- $f''(x) = 0$: possibly a point of inflection — where concavity changes from up to down or vice versa.

In physics, if y is displacement, then dy/dx is velocity and d^2y/dx^2 is the second derivative with respect to distance. When x is time t: dy/dt = velocity and d^2y/dt^2 = acceleration.

Worked Example: Computing Second Derivatives

Find $f''(x)$ for: (i) $f(x) = x^4 - 3x^2 + 2$ (ii) $f(x) = \sin(2x)$ (iii) $f(x) = \ln(x^2 + 1)$

Step 1: $f(x) = x^4 - 3x^2 + 2$

$f'(x) = 4x^3 - 6x$

$f''(x) = 12x^2 - 6$

Step 2: $f(x) = \sin(2x)$

$f'(x) = 2\cos(2x)$ (chain rule)

$f''(x) = -4\sin(2x)$ (chain rule again)

Step 3: $f(x) = \ln(x^2 + 1)$

$f'(x) = 2x/(x^2 + 1)$ (chain rule)

$f''(x) = d/dx[2x/(x^2 + 1)]$. Apply quotient rule: $u = 2x$, $v = x^2+1$.

$f''(x) = [2(x^2+1) - 2x\cdot 2x] / (x^2+1)^2 = [2x^2+2-4x^2]/(x^2+1)^2 = (2-2x^2)/(x^2+1)^2$

3.12 nth-Order Derivatives and the Leibnitz Theorem

The n-th derivative of a function is defined recursively as the derivative of the (n−1)-th derivative. For specific function families, there are elegant closed-form expressions for the n-th derivative. These are particularly useful in Taylor/Maclaurin series expansions and in differential equation solutions.

Standard nth Derivatives

Function	nth Derivative $D^n y$
$y = x^m$ $(m \geq n)$	$D^n(x^m) = m(m-1)(m-2)\cdots(m-n+1) \cdot x^{(m-n)} = m!/(m-n)! \cdot x^{(m-n)}$
$y = e^x$	$D^n(e^x) = e^x$
$y = e^{ax}$	$D^n(e^{ax}) = a^n \cdot e^{ax}$
$y = \ln x$	$D^n(\ln x) = (-1)^{(n-1)} \cdot (n-1)! / x^n$ for $n \geq 1$
$y = \sin x$	$D^n(\sin x) = \sin(x + n\pi/2)$
$y = \cos x$	$D^n(\cos x) = \cos(x + n\pi/2)$
$y = \sin(ax + b)$	$D^n = a^n \cdot \sin(ax + b + n\pi/2)$
$y = (ax + b)^m$	$D^n = a^n \cdot m!/(m-n)! \cdot (ax + b)^{(m-n)}$ for $m \geq n$

Leibnitz's Theorem

Leibnitz's Theorem provides a formula for the n-th derivative of the product of two functions — the generalisation of the product rule to any order. If u and v are both differentiable n times, then:

Leibnitz's Theorem: $D^n(uv) = \Sigma(r=0 \text{ to } n)\ C(n,r) \cdot D^{n-r}(u) \cdot D^r(v)$

where $C(n,r) = n! / [r!(n-r)!]$ is the binomial coefficient. Written explicitly:

Expanded form: $D^n(uv) = uD^n v + C(n,1)Du \cdot D^{n-1}v + C(n,2)D^2u \cdot D^{n-2}v + \ldots + D^n u \cdot v$

This formula is the exact analogue of the binomial expansion $(a+b)^n$ applied to differentiation operators. Choose u to be the function whose derivatives eventually become zero (polynomials) or periodic (sin/cos), and v to be the other.

📝 Worked Example: Leibnitz Theorem — Finding nth Derivative

Find the nth derivative of $y = x^2 \cdot e^x$ using Leibnitz's Theorem.

Step 1: Let $u = x^2$ and $v = e^x$. Choose $u = x^2$ (polynomial — becomes zero after 3rd derivative) and $v = e^x$.

Step 2: Compute the required derivatives of u:

$Du = 2x, \quad D^2u = 2, \quad D^3u = 0$ (and all higher derivatives of u are 0)

Step 3: Compute derivatives of $v = e^x$: $D^r(e^x) = e^x$ for all r.

Step 4: Apply Leibnitz's Theorem (note all terms with D^3u and beyond vanish):

$$D^n(x^2 \cdot e^x) = x^2 \cdot e^x + C(n,1) \cdot 2x \cdot e^x + C(n,2) \cdot 2 \cdot e^x + 0 + \ldots$$

$$= e^x[x^2 + n(2x) + n(n-1)/2 \cdot 2]$$

$$= e^x[x^2 + 2nx + n(n-1)]$$

📝 Worked Example: Leibnitz Theorem — y = x sin x

Find $D^n(x \sin x)$.

Step 1: Let $u = x$, $v = \sin x$. $D^0u = x$, $Du = 1$, $D^2u = 0$ (and beyond). $D^r(\sin x) = \sin(x + r\pi/2)$.

Step 2: Apply Leibnitz (only first two terms are non-zero since $D^2u = 0$):

$$D^n(x \sin x) = x \cdot D^n(\sin x) + C(n,1) \cdot 1 \cdot D^{n-1}(\sin x)$$

$$= x \cdot \sin(x + n\pi/2) + n \cdot \sin(x + (n-1)\pi/2)$$

3.13 Angle Between Two Curves

When two curves intersect at a point, they generally meet at an angle. The angle between two curves at a point of intersection is defined as the angle between their tangent lines at that point. Since the tangent line at a point has slope equal to dy/dx at that point, finding the angle between two curves reduces to a problem in coordinate geometry.

Formula for the Angle of Intersection

Let the two curves have slopes m_1 and m_2 at their point of intersection (i.e., $m_1 = dy/dx$ for curve 1 and $m_2 = dy/dx$ for curve 2, evaluated at the intersection point). The angle θ between the tangent lines satisfies:

Angle between curves: $\tan\theta = \left| (m_1 - m_2) / (1 + m_1 m_2) \right|$

Here, the absolute value ensures we take the acute angle ($0 \le \theta \le 90°$). The formula gives $\theta = 90°$ (orthogonal curves) if $m_1 m_2 = -1$, and $\theta = 0°$ (tangent curves) if $m_1 = m_2$.

Orthogonal Curves

Two curves are orthogonal at a point of intersection if their tangent lines are perpendicular at that point, i.e., $m_1 \cdot m_2 = -1$. Orthogonal curve families appear naturally in physics: electric field lines and equipotential surfaces, fluid streamlines and equipressure curves.

Worked Example: Angle of Intersection Between Two Parabolas

Find the angle of intersection between $y = x^2$ and $y^2 = x$.

Step 1: Find the intersection points: substitute $y = x^2$ into $y^2 = x$.

$(x^2)^2 = x \implies x^4 - x = 0 \implies x(x^3 - 1) = 0 \implies x = 0$ or $x = 1$

Intersection points: (0, 0) and (1, 1).

Step 2: Find slopes at (1, 1):

Curve 1: $y = x^2 \implies dy/dx = 2x$. At (1,1): $m_1 = 2$.

Curve 2: $y^2 = x \implies 2y(dy/dx) = 1 \implies dy/dx = 1/(2y)$. At (1,1): $m_2 = 1/2$.

Step 3: Apply the angle formula:

$\tan\theta = |(m_1 - m_2)/(1 + m_1m_2)| = |(2 - 1/2)/(1 + 2\cdot 1/2)| = |(3/2)/(2)| = 3/4$

$\theta = \tan^{-1}(3/4) \approx 36.87°$

At (0,0): $m_1 = 0$, $m_2 \to \infty$ (vertical tangent), so curves are perpendicular ($\theta = 90°$).

Analysing Misconceptions: Continuity and Differentiability Are Equivalent

✗ MYTH: If a function is continuous at a point, it must be differentiable there as well.

✓ REALITY: Continuity is a necessary condition for differentiability, but not sufficient. The classic counterexample is $f(x) = |x|$: it is continuous at $x = 0$ (no break), but not differentiable there (the graph has a sharp corner, so the slope from the left is -1 and from the right is $+1$ — they disagree). Geometrically, differentiability requires a smooth tangent line, while continuity only requires no break in the curve.

3.14 Polar Coordinates and Polar Curves

Polar coordinates offer an alternative to the Cartesian (x, y) system. Instead of specifying a point's horizontal and vertical distances from the origin, polar coordinates specify its distance from the origin and the angle it makes with the positive x-axis. This coordinate system is natural for curves that have rotational symmetry — spirals, circles centred at the origin, cardioids, roses, and lemniscates.

The Polar Coordinate System

A point P in the plane is described in polar coordinates as (r, θ), where r = distance from the origin O to P (the radial distance) and θ = angle from the positive x-axis to OP, measured counterclockwise (in radians).

The conversion between polar and Cartesian coordinates is given by:

Polar to Cartesian: $x = r \cdot \cos\theta, \quad y = r \cdot \sin\theta$

Cartesian to Polar: $r = \sqrt{x^2 + y^2}, \quad \tan\theta = y/x$

Important Standard Polar Curves

Curve	Equation	Description
Circle at origin	$r = a$	Centred at origin, radius a
Cardioid	$r = a(1 \pm \cos\theta)$ or $r = a(1 \pm \sin\theta)$	Heart-shaped; max $r = 2a$, passes through origin
Rose Curve	$r = a\cdot\cos(n\theta)$ or $r = a\cdot\sin(n\theta)$	n-petal if n is odd; 2n-petal if n is even
Lemniscate	$r^2 = a^2\cdot\cos(2\theta)$ or $r^2 = a^2\cdot\sin(2\theta)$	Figure-eight shape
Archimedean Spiral	$r = a\theta$	Constant increase in r per revolution
Limaçon	$r = a + b\cdot\cos\theta$	Inner loop if $b > a$

📝 Worked Example: Sketching a Polar Curve — Cardioid

Trace the cardioid $r = a(1 + \cos\theta)$ for key values of θ.

Step 1: Compute r at key angles:

$\theta = 0$: $r = a(1+1) = 2a$ (maximum, rightmost point)

$\theta = \pi/3$: $r = a(1+1/2) = 3a/2$

$\theta = \pi/2$: $r = a(1+0) = a$ (top of curve)

$\theta = \pi$: $r = a(1-1) = 0$ (curve passes through origin)

$\theta = 3\pi/2$: $r = a(1+0) = a$ (bottom of curve)

Step 2: By symmetry about the x-axis (cosine is even), the curve is symmetric.

Step 3: The curve completes one full petal as θ goes from 0 to 2π.

Area enclosed by cardioid = $(1/2)\int_0^{2\pi} r^2\, d\theta = (3/2)\pi a^2$ (computed using integral calculus).

Did You Know?

The cardioid curve — $r = a(1 + \cos\theta)$ in polar form — gets its name from the Greek word for heart (kardia). The antenna gain patterns of directional microphones and certain radio antennas follow a cardioid shape: they are most sensitive in one direction and insensitive from behind. Microphones used by news anchors on Indian news channels such as NDTV and ABP News use cardioid condenser microphones to pick up sound from the speaker's direction while rejecting ambient studio noise — a direct application of polar curve geometry in engineering design.

3.15 Differentiation in Polar Coordinates and the Pedal Equation

Having established polar coordinates and standard polar curves, we now develop the calculus of polar functions. The key geometric concept is the angle φ (phi) between the radius vector OP and the tangent to the curve at P. This angle, together with the polar derivative, gives us complete geometric information about how a polar curve bends and turns.

The Angle φ Between the Radius Vector and Tangent

For a polar curve $r = f(\theta)$, the angle φ between the radius vector (from origin to the point) and the tangent to the curve at that point satisfies:

Angle between radius and tangent: $\tan\varphi = r / (dr/d\theta) = r / r'$

where $r' = dr/d\theta$. This is a fundamental formula in polar geometry. When $\varphi = \pi/2$, the tangent is perpendicular to the radius vector (the curve is moving "radially"). When $\varphi = \pi/4$, the curve meets the radius at 45°.

The Pedal Equation

The pedal equation of a curve relates the perpendicular distance p from the origin (the pole) to the tangent at any point, to the radial distance r from the origin to that point. It provides a relationship $p = f(r)$ that is intrinsic to the curve — independent of the choice of orientation.

The perpendicular distance p from the pole to the tangent of a polar curve $r = f(\theta)$ is given by:

Pedal equation formula: $p = r \cdot \sin\varphi = r^2 / \sqrt{(r^2 + r'^2)}$

where $r' = dr/d\theta$. The relationship between p and r obtained by eliminating θ is the pedal equation.

Worked Example: Angle φ and Pedal Equation for an Archimedean Spiral

For the Archimedean spiral $r = a\theta$, find: (i) $\tan\varphi$, (ii) the pedal equation.

Step 1: Compute $r' = dr/d\theta = a$.

Step 2: $\tan\varphi = r/r' = a\theta/a = \theta$. So $\varphi = \tan^{-1}(\theta)$.

As θ increases, φ increases — the tangent becomes more perpendicular to the radius.

Step 3: For the pedal equation: $r' = a$, so $r^2 + r'^2 = a^2\theta^2 + a^2 = a^2(\theta^2 + 1) = a^2(r^2/a^2 + 1) = r^2 + a^2$.

$p = r^2/\sqrt{(r^2 + a^2)}$

Therefore the pedal equation is: $p^2 = r^4/(r^2 + a^2)$, or equivalently $r^2(r^2 + a^2 - r^2) = a^2r^2/p^2 \cdot p^2 \to p^2(r^2 + a^2) = r^4$.

Pedal equation: $p(r^2 + a^2)$^(1/2) $= r^2$, or $p = r^2/\sqrt{(r^2 + a^2)}$.

Worked Example: Pedal Equation of the Cardioid

For the cardioid $r = a(1 + \cos\theta)$, find the pedal equation.

Step 1: $r' = dr/d\theta = -a\sin\theta$

Step 2: $r^2 + r'^2 = a^2(1+\cos\theta)^2 + a^2\sin^2\theta$

$= a^2[(1+\cos\theta)^2 + \sin^2\theta]$

$= a^2[1 + 2\cos\theta + \cos^2\theta + \sin^2\theta]$

$= a^2[2 + 2\cos\theta] = 2a^2(1 + \cos\theta) = 2ar$

Step 3: $p = r^2/\sqrt{(r^2 + r'^2)} = r^2/\sqrt{(2ar)} = r^2/(\sqrt{(2a)}\cdot\sqrt{r}) = r^{(3/2)}/\sqrt{(2a)}$

Step 4: Square both sides: $p^2 = r^3/(2a)$

Pedal equation of the cardioid: $2ap^2 = r^3$

Let's Check

Q.No	Type	Question
1	MCQ	The angle between two curves is found using the tangent slopes m_1 and m_2 at the intersection point via:
2	MCQ	For the polar curve $r = a(1 + \cos\theta)$, what is the value of r at $\theta = \pi$?
3	Fill in Blank	In polar coordinates, the angle φ between the radius vector and the tangent satisfies $\tan\varphi$ = ______.
4	Fill in Blank	Two curves are orthogonal if the product of their slopes at the intersection equals ______.

5	True/False	The pedal equation relates the perpendicular distance p from the pole to the tangent, with the radial distance r.
6	True/False	The n-th derivative of e^x for any n is always e^x.

Answers:

Q.No	Answer
1	b) $\tan\theta = \lvert(m_1 - m_2)/(1 + m_1m_2)\rvert$
2	b) 0 — $r = a(1 + \cos\pi) = a(1-1) = 0$. The cardioid passes through the origin at $\theta = \pi$.
3	$r / (dr/d\theta)$ or r / r'
4	-1
5	True — that is the precise definition of the pedal equation.
6	True — $D^n(e^x) = e^x$ for all $n \geq 0$.

📘 Case Study: Derivatives in Action — Optimising the Mumbai-Pune Expressway Curve Design

Background:

The Mumbai-Pune Expressway, opened in 2002 and managed by MSRDC (Maharashtra State Road Development Corporation), was India's first six-lane controlled-access expressway. The 94.6-km route traverses the Western Ghats, requiring careful engineering of curves, gradients, and transitions at various sections. The mathematics of differential calculus — specifically derivatives and curvature — is central to every aspect of highway curve design.

The Engineering Problem:

When a vehicle travels along a curved section of road, several forces act on it: centripetal force (requiring the road to curve), gravity (requiring superelevation — banking the road), and friction. At any point on the road, the curvature κ (kappa) determines the required centripetal force:

Curvature formula: $\kappa = |d^2y/dx^2| / [1 + (dy/dx)^2]^{(3/2)}$

For a road segment $y = f(x)$, the curvature κ uses both the first and second derivatives. The radius of curvature $R = 1/\kappa$ determines the minimum speed at which a vehicle can safely negotiate the curve without sliding. MSRDC engineers use this formula to specify the superelevation (banking angle) at each curve.

Parametric Curves in Highway Design:

Transition curves — the smooth sections between a straight road and a circular curve — are designed as Euler spirals (also called clothoids), where the curvature increases linearly with arc length. This is a parametric curve. Engineers compute $dy/dx = (dy/ds)/(dx/ds)$ and d^2y/dx^2 in parametric form exactly as studied in Sub-unit 3.10, to verify that the curvature transitions are smooth (no sudden change that would cause passenger discomfort or vehicle instability).

Continuity and Differentiability Requirements:

Highway alignment specifications from the Indian Roads Congress (IRC) require that the road profile y(x) must be at least C^2 continuous — twice differentiable — at every junction between road segments. A C^0 discontinuity would be a crack in the road. A C^1 discontinuity would be a kink (sharp corner) — dangerous. A C^2 discontinuity would cause a sudden jerk (abrupt change in acceleration) — uncomfortable and potentially dangerous at speed. The mathematical concepts of continuity and differentiability studied in Sub-units 3.1 and 3.2 are direct engineering requirements in road design.

Optimisation Using Derivatives:

The expressway passes through hilly terrain where the gradient (dy/dx, the first derivative) must be minimised to reduce fuel consumption and braking requirements. IRC guidelines specify a maximum gradient of 5% ($dy/dx \leq 0.05$). Where the gradient exceeds this, tunnels (like the Katraj Tunnel) or elevated sections are required. Finding the optimal routing that minimises the maximum gradient is an optimisation problem — solved by setting the derivative equal to zero and analysing critical points, exactly as studied in this module.

(Source: Maharashtra State Road Development Corporation technical reports; Indian Roads Congress guidelines IRC:SP:23-1993.)

Module Summary

This module has built a complete and systematic framework for differential calculus — from the foundational concept of continuity through the full range of differentiation techniques and their elegant geometric applications.

Continuity at a point requires three conditions: the function value exists, the limit exists, and they are equal. Differentiability implies continuity, but the converse is false — continuous functions may have corners or cusps where no derivative exists. The derivative from first principles, defined as a limit of the difference quotient, gives the instantaneous rate of change and geometrically represents the slope of the tangent line.

The differentiation toolkit built in this module includes:

- Power rule, sum/difference rule, constant multiple rule — for algebraic functions.
- Trigonometric derivatives — $\sin x \to \cos x$, $\cos x \to -\sin x$, and four more standard results.
- Exponential and logarithmic derivatives — $d/dx(e^x) = e^x$, $d/dx(\ln x) = 1/x$, and logarithmic differentiation for variable exponents.
- Chain rule — $d/dx[f(g(x))] = f'(g(x))\cdot g'(x)$ — the most important rule for composite functions.
- Product and quotient rules for products and ratios of functions.
- Derivatives of all six inverse trigonometric functions — expressed in terms of algebraic functions.
- Implicit differentiation — for curves defined by $F(x,y) = 0$.
- Parametric differentiation — $dy/dx = (dy/dt)/(dx/dt)$; second derivative via $d/dt(dy/dx)/(dx/dt)$.

Higher-order differentiation extends to second derivatives (curvature and concavity) and n-th derivatives. The Leibnitz Theorem $D^n(uv) = \Sigma\, C(n,r)\cdot D^{n-r}(u)\cdot D^r(v)$ provides an elegant closed-form formula for the n-th derivative of any product, with special cases yielding beautiful results for $x^n e^x$, $x^n \sin x$, etc.

The geometric applications — angle between curves using $\tan\theta = |(m_1-m_2)/(1+m_1m_2)|$, polar curves and their differentiation, and the pedal equation $p = r^2/\sqrt{(r^2+r'^2)}$ — demonstrate that differentiation is not merely a computational tool but a lens for understanding geometric properties of curves. Polar coordinates provide a natural setting for curves with rotational symmetry, and the polar derivative formula $\tan\varphi = r/r'$ captures the intrinsic geometric character of each polar curve.

Glossary

Term	Definition
Continuity at a Point	f(x) is continuous at x = a if: (1) f(a) exists, (2) lim[x→a] f(x) exists, (3) lim[x→a] f(x) = f(a). All three must hold.
Derivative	The instantaneous rate of change of a function: f'(x) = lim[h→0] [f(x+h)−f(x)]/h. Geometrically, the slope of the tangent line.
Chain Rule	d/dx[f(g(x))] = f'(g(x))·g'(x). Used for composite functions — multiply derivative of outer by derivative of inner.
Implicit Differentiation	Differentiating both sides of F(x,y) = 0 w.r.t. x, treating y as a function of x and applying the chain rule when differentiating y.
Parametric Differentiation	For x = f(t), y = g(t): dy/dx = (dy/dt)/(dx/dt). The curve is described via a parameter t.
Leibnitz's Theorem	$D^n(uv) = \Sigma(r=0 \text{ to } n)\ C(n,r)\cdot D^{n-r}u\cdot D^r v$. Gives the n-th derivative of a product in terms of lower derivatives of each factor.
Second Derivative	d^2y/dx^2 — the derivative of dy/dx. Positive means concave up; negative means concave down. Related to curvature of the curve.
Logarithmic Differentiation	Technique for differentiating y = [u(x)]^v(x): take ln of both sides, differentiate implicitly, then solve for dy/dx.
Pedal Equation	The relationship p = f(r) between the perpendicular distance p from the pole to the tangent and the radial distance r for a polar curve.
Polar Coordinates	System (r, θ) where r = distance from origin and θ = angle from positive x-axis. Related to Cartesian by x = r cos θ, y = r sin θ.

Self Assessment Questions

Section A: Multiple Choice Questions (MCQ)

Choose the most appropriate answer for each question.

1. A function f(x) is continuous at x = a if:

   a) f(a) exists

   b) lim[x→a] f(x) = f(a) and both exist

   c) f(a) is defined only

   d) The derivative exists at a

2. The derivative of f(x) from first principles is:

   a) [f(x+h)+f(x)]/h as h→0

   b) [f(x+h)−f(x)]/h as h→0

   c) f(x+h)·f(x)/h

   d) [f(x)−f(x+h)]/h as h→∞

3. d/dx(x^n) equals:

   a) nx^(n+1)

   b) x^{n-1}

   c) nx^{n-1}

   d) x/n

4. The derivative of e^(3x+2) is:

   a) e^(3x+2)

   b) 3e^(3x+2)

   c) (3x+2)e^(3x+1)

   d) 3e^(3x)

5. d/dx(tan x) equals:

   a) sec x

   b) $-\text{cosec}^2 x$

c) $\sec^2 x$

d) sec x tan x

6. The chain rule for y = f(g(x)) states dy/dx equals:

a) $f'(x) \cdot g'(x)$

b) $f'(g(x)) \cdot g'(x)$

c) $f(g'(x))$

d) $f'(x)/g'(x)$

7. $d/dx(\sin^{-1} x)$ is:

a) $-1/\sqrt{(1-x^2)}$

b) $1/(1+x^2)$

c) $1/\sqrt{(1-x^2)}$

d) $1/(x\sqrt{(x^2-1)})$

8. For parametric equations $x = t^2$, $y = t^3$, dy/dx equals:

a) 3t/2

b) t/2

c) $2t/3t^2$

d) $3t^2/2t$

9. If f(x) is differentiable at x = a, then:

a) f may be discontinuous at a

b) f must be continuous at a

c) f'(a) = 0

d) f is constant near a

10. d/dx(ln x) equals:

a) ln x

b) x

c) 1/x

d) e^x

11. The second derivative test: if f''(a) > 0 at a critical point, then x = a is:

a) A local maximum

b) A local minimum

c) A point of inflection

d) A discontinuity

12. The Leibnitz Theorem for the nth derivative of a product uv uses:

a) Product rule applied n times

b) Binomial coefficients C(n,r)

c) Quotient rule n times

d) Chain rule only

13. The angle θ between two curves with slopes m_1 and m_2 satisfies:

a) $\tan \theta = m_1 - m_2$

b) $\tan \theta = m_1 \cdot m_2$

c) $\tan \theta = |(m_1 - m_2)/(1 + m_1 m_2)|$

d) $\tan \theta = (m_1 + m_2)/(1 - m_1 m_2)$

14. In polar coordinates, r = a(1 + cos θ) represents a:

a) Circle

b) Spiral

c) Lemniscate

d) Cardioid

15. For a polar curve r = f(θ), tan φ (where φ is the angle between radius and tangent) equals:

a) r'/(r)

b) r/r' where r' = dr/dθ

c) r·r'

d) $1/r'$

16. The pedal equation gives a relation between:

a) x and y

b) r and θ

c) p (perpendicular from pole to tangent) and r

d) Slope m and intercept c

17. For $y = x^x$, logarithmic differentiation gives dy/dx as:

a) x^{x-1}

b) $x^x \cdot \ln x$

c) $x^x(1 + \ln x)$

d) $x^x/\ln x$

18. If $f(x) = \sin(ax+b)$, then $D^n(f)$ equals:

a) $a^n \cdot \sin(ax+b)$

b) $\sin(ax+b+n\pi/2)$

c) $a^n \cdot \sin(ax+b+n\pi/2)$

d) $n \cdot \sin(ax+b)$

19. Two curves are orthogonal at their intersection if:

a) $m_1 = m_2$

b) $m_1 + m_2 = 0$

c) $m_1 \cdot m_2 = -1$

d) $m_1 \cdot m_2 = 1$

20. The type of discontinuity where lim exists but ≠ f(a) (or f(a) undefined) is called:

a) Jump discontinuity

b) Infinite discontinuity

c) Oscillatory discontinuity

d) Removable discontinuity

Section B: True / False

State whether each statement is True or False.

1. Differentiability implies continuity, but continuity does not imply differentiability.

2. The derivative of cos x is sin x.

3. For implicit differentiation of $x^3 + y^3 = 6xy$, when we differentiate y^3 w.r.t. x, we get $3y^2$ (without any dy/dx factor).

4. The pedal equation $p = r^2/\sqrt{(r^2+r'^2)}$ relates the perpendicular from the origin to the tangent with the radial distance.

5. $d/dx(\tan^{-1}x) = 1/\sqrt{(1-x^2)}$.

Section C: Short Answer / Problem Solving

Answer each question with full working and explanation.

1. Find the derivatives of: (i) $y = (x^2 + 1)$^(1/2) · sin(3x), (ii) y = e^(x^2)·ln(2x), (iii) $y = \tan^{-1}(\sqrt{x})$. Show each differentiation rule applied.

2. If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, find dy/dx and d^2y/dx^2. Simplify your answers.

3. Find the nth derivative of $y = x^2 \cdot \sin x$ using Leibnitz's Theorem. Identify which terms vanish and write the final compact form.

4. Find the angle of intersection between the curves $x^2 + y^2 = 4$ (circle) and $x^2 - y^2 = 2$ (rectangular hyperbola). State whether the curves are orthogonal.

5. For the cardioid $r = a(1 + \cos \theta)$: (i) Find $dr/d\theta$ and compute $\tan \varphi$ at $\theta = \pi/3$. (ii) Derive the pedal equation and show it equals $2ap^2 = r^3$.

Self Assessment — Answer Keys

Section A — MCQ

Q.No	Answer	Option Text
1	**B**	b) lim[x→a] f(x) = f(a) and both exist
2	**B**	b) [f(x+h)−f(x)]/h as h→0
3	**C**	c) nx^{n-1}
4	**B**	b) 3e^(3x+2)
5	**C**	c) $\sec^2 x$
6	**B**	b) f'(g(x))·g'(x)
7	**C**	c) $1/\sqrt{(1-x^2)}$
8	**A**	a) 3t/2
9	**B**	b) f must be continuous at a
10	**C**	c) 1/x
11	**B**	b) A local minimum
12	**B**	b) Binomial coefficients C(n,r)
13	**C**	c) $\tan\theta = \|(m_1-m_2)/(1+m_1m_2)\|$
14	**D**	d) Cardioid
15	**B**	b) r/r' where r' = dr/dθ
16	**C**	c) p (perpendicular from pole to tangent) and r
17	**C**	c) $x^x(1 + \ln x)$
18	**C**	c) $a^n \cdot \sin(ax+b+n\pi/2)$

19	**C**	c) $m_1 \cdot m_2 = -1$
20	**D**	d) Removable discontinuity

Section B — True / False Answer Key

Q.No	Answer	Notes
1	**True**	Correct.
2	**False**	False — $d/dx(\cos x) = -\sin x$ (negative sign).
3	**False**	False — differentiating y^3 w.r.t. x gives $3y^2 \cdot (dy/dx)$ by the chain rule.
4	**True**	Correct.
5	**False**	False — $d/dx(\tan^{-1}x) = 1/(1+x^2)$, not $1/\sqrt{(1-x^2)}$.

Section C — Short Answer Model Solutions

Q.No	Model Solution
1	(i) $y = \sqrt{(x^2+1)} \cdot \sin(3x)$. Product rule: $dy/dx = [x/\sqrt{(x^2+1)}] \cdot \sin(3x) + \sqrt{(x^2+1)} \cdot 3\cos(3x)$. (ii) $y = e^{(x^2)} \cdot \ln(2x)$. $dy/dx = 2x \cdot e^{(x^2)} \cdot \ln(2x) + e^{(x^2)} \cdot (1/x) = e^{(x^2)}[2x \cdot \ln(2x) + 1/x]$. (iii) $y = \tan^{-1}(\sqrt{x})$. Chain rule: $dy/dx = 1/(1+(\sqrt{x})^2) \cdot (1/2)x^{(-1/2)} = 1/(2\sqrt{x}(1+x))$.
2	$dx/dt = a(-\sin t + \sin t + t \cos t) = at \cos t$. $dy/dt = a(\cos t - \cos t + t \sin t) = at \sin t$. $dy/dx = (at \sin t)/(at \cos t) = \tan t$. For d^2y/dx^2: $d/dt(\tan t) = \sec^2 t$. $d^2y/dx^2 = \sec^2 t/(at \cos t) = 1/(at \cos^3 t) = \sec^3 t/(at)$.
3	Let $u = x^2$, $v = \sin x$. $Du = 2x$, $D^2u = 2$, $D^n u = 0$ for $n \geq 3$. $D^r(\sin x) = \sin(x+r\pi/2)$. Leibnitz: $D^n(x^2 \sin x) = x^2 \cdot \sin(x+n\pi/2) +$

	$C(n,1)\cdot 2x\cdot\sin(x+(n-1)\pi/2) + C(n,2)\cdot 2\cdot\sin(x+(n-2)\pi/2) = x^2\sin(x+n\pi/2) + 2nx\cdot\sin(x+(n-1)\pi/2) + n(n-1)\sin(x+(n-2)\pi/2)$.
4	Intersection: $x^2+y^2=4$, $x^2-y^2=2$. Adding: $2x^2=6 \rightarrow x^2=3 \rightarrow x=\pm\sqrt{3}$. Then $y^2=1 \rightarrow y=\pm 1$. Points: $(\pm\sqrt{3}, \pm 1)$. Slopes: Curve 1: $2x+2y(dy/dx)=0 \rightarrow m_1=-x/y$. Curve 2: $2x-2y(dy/dx)=0 \rightarrow m_2=x/y$. At $(\sqrt{3},1)$: $m_1=-\sqrt{3}$, $m_2=\sqrt{3}$. $m_1\cdot m_2=-3 \neq -1$, so NOT orthogonal. Angle: $\tan\theta=\lvert(-\sqrt{3}-\sqrt{3})/(1+(-\sqrt{3})(\sqrt{3}))\rvert=\lvert-2\sqrt{3}/(1-3)\rvert=\lvert-2\sqrt{3}/(-2)\rvert=\sqrt{3}$. $\theta=60°$.
5	(i) $r = a(1+\cos\theta)$, $dr/d\theta = -a\sin\theta$. $\tan\varphi = r/(dr/d\theta) = a(1+\cos\theta)/(-a\sin\theta)$. At $\theta=\pi/3$: $\tan\varphi = a(1+1/2)/(-a\cdot\sqrt{3}/2) = (3/2)/(-\sqrt{3}/2) = -\sqrt{3}$. $\varphi = 120°$ (measured from positive r direction). (ii) $r^2+(dr/d\theta)^2 = a^2(1+\cos\theta)^2 + a^2\sin^2\theta = a^2(2+2\cos\theta) = 2ar$. So $p = r^2/\sqrt{(2ar)} = r^{(3/2)}/\sqrt{(2a)}$. Squaring: $p^2 = r^3/(2a)$. Therefore $2ap^2 = r^3$. QED.

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